The Empirical Case for Gaussian Renormalization as Default

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Provisional draft, 17 August 2026

Abstract

Given probabilities over mutually exclusive events, and told that one has been *withdrawn*, scratched, delisted, masked out of a softmax, proportional renormalization is the reflex. Conditioning is available but not free: it needs a model of how the probabilities arose, and renormalization *tacitly* supplies one. By Yellott's theorem it is the model in which they came from a contest with Gumbel noise. Assume Gaussian noise instead and the survivors' odds contract by a computable amount. Both assumptions are calibrated to the same probabilities and fit nothing, so they can be scored against each other. The Gaussian contest forecasts better on an experiment where subjects chose from all thirty-one subsets of five goods, and on twelve of twelve ranking collections. Contraction of a noisy estimate lowers log loss by itself, so we check against populations that renormalize exactly; the advantage survives except in the smallest collection.

1 Two restriction maps

For a population and a menu S, let p_i be the share choosing or first-ranking item i. Luce's axiom [5] assigns a worth $u(i)$ with

$$p_i = \frac{u(i)}{\sum_{j \in S} u(j)}, \tag{1}$$

so the shares determine the worths up to scale. The prediction for a subset $T \subseteq S$ follows by changing the denominator alone, which leaves every ratio p_i/p_j with $i, j \in T$ exactly as it was in S. For a two-item menu the predicted share preferring i is therefore $p_i/(p_i + p_j)$. This invariance is Independence of Irrelevant Alternatives.

The reflex to renormalize is not confined to choice modelling, and it is strong enough to be mistaken for a theorem. Renormalization is the posterior for mutually exclusive events only when the information that one is impossible is uninformative about which of the rest obtained; Monty Hall is the familiar counterexample, and the belief that renormalization is automatically Bayes survives repeated exposure to it. Withdrawal is a third case, neither flat-likelihood conditioning nor informative conditioning, because no information about a fixed experiment has arrived: an alternative is gone and the problem is a different one. Conditioning is still available, but it now requires a model of how the probabilities were generated, and renormalization is one such model rather than a consequence of probability theory. By Yellott's theorem it is the model with Gumbel noise. Which generative assumption to make is then an empirical question, and this paper measures two of them.

Thurstone [9] connects the same two quantities through a contest. Item i carries a location a_i, draws a performance $X_i \sim N(a_i, 1)$ on each occasion, and the highest draw is chosen, so

$$p_i = \int \varphi(x - a_i) \prod_{k \neq i} \Phi(x - a_k)\, dx. \tag{2}$$

The integral is one-dimensional for any number of alternatives, since conditioning on $X_i = x$ makes the other events independent, and it is inverted numerically by a lattice method [2]; locations are identified up to a common translation, fixed here by $\sum_i a_i = 0$. Prediction for a subset T means re-running the contest among T alone. For a pair this is one normal evaluation,

$$P(X_i > X_j) = \Phi\left(\frac{a_i - a_j}{\sqrt{2}}\right), \tag{3}$$

and if $p_i > p_j$ it lies between $\frac{1}{2}$ and $p_i/(p_i + p_j)$: removing the other alternatives contracts the odds toward one. Winning a large field requires an unusually high draw, and beating a single rival does not, so the favourite's advantage counts for less. Proposition 1 makes this exact.

Both rules are calibrated to the same $|S|-1$ free shares, and neither sees any observation from a smaller menu. The comparison is between two maps with no free parameters left, rather than between two fitted models. This also fixes what the comparison can settle. A fitted Plackett–Luce likelihood uses the whole ranking, and a fitted Bradley–Terry model uses the pairwise outcomes directly; neither is estimated here. The question is which map from full-menu shares to restricted-menu shares predicts population choice better. Both are *representative-agent* maps: each treats the population as one chooser with one set of worths or locations, and transports its shares from S to T.

The direction of the disagreement is a theorem, and it holds for noise laws other than the Gaussian. Let f and F be the density and distribution of the common noise and let $r = f/F$ be its reverse hazard, with the highest draw winning as above.

Proposition 1 (Removing alternatives contracts the odds)**.** *Suppose* $\log r$ *is concave. Let* $T \subset S$ *and* $i, j \in T$ *with* $a_i > a_j$*. Then*

$$\frac{P(i \mid S)}{P(j \mid S)} \;\geq\; \frac{P(i \mid T)}{P(j \mid T)} \;\geq\; 1,$$

with the first inequality strict when $\log r$ *is strictly concave and* $S \setminus T$ *is non-empty.*

Proof. Write $A(x) = f(x-a_i)F(x-a_j)$, $B(x) = f(x-a_j)F(x-a_i)$ and $H(x) = \prod_{k \in S \setminus T} F(x-a_k)$, so that $P(i \mid T) \propto \int A$ and $P(i \mid S) \propto \int AH$, and likewise for j with B. Then $A(x)/B(x) = r(x-a_i)/r(x-a_j)$, which is increasing in x when $\log r$ is concave, since $a_i > a_j$ places $x-a_i$ below $x - a_j$ and $(\log r)'$ is decreasing. Under the probability measure proportional to $B(x)\, dx$ the two functions A/B and H are both increasing, hence positively correlated, so $\int AH / \int BH \geq \int A / \int B$. □

The proposition says that removing alternatives moves the ratio toward one.

Two laws sit at the ends of that condition. The Gaussian satisfies it strictly, since $\frac{d^2}{dx^2} \log r(x) = -\operatorname{Var}(Z \mid Z < x) < 0$. The Gumbel has $r(x) = s^{-1}e^{-x/s}$, so $\log r$ is affine, the inequality becomes an equality, and renormalization is recovered exactly. That is the sufficiency half of Yellott's theorem [10]. The axiom is a boundary of the class, not a rival to it.

The boundary is where machine learning sits. A softmax over scores z_i is Equation (1) with $u(i) = \exp(z_i)$, so masking a softmax and rescaling the survivors is proportional renormalization, and by Yellott's theorem that is the Gumbel-noise member of the contest class. The default is a single point of the family.

The proposition gives the direction of the effect, not its size. Size depends on the shares as well as on the noise law: recalibrating several laws to one share vector and then to another can reverse which of them contracts more. We therefore make no claim that thinner or heavier tails contract more.

2 Choices from real menus

Twelve of the thirteen collections used here are complete rankings, from which subset choice is read off rather than observed. One is not, and it comes first.

Costa-Gomes, Cueva, Gerasimou and Tejiscak [1] had each subject choose separately from all thirty-one subsets of five consumer goods, so restriction is observed rather than imputed. The archive names the goods only by two-letter codes and we describe them no more precisely. This is the cleanest available test of the question above, and it is the primary result.

Subjects are split into five folds. On the training subjects, shares come only from the full five-item menu; both rules are calibrated to those shares and nothing else; each then predicts every menu of size two, three and four; and we score the choices actually made by held-out subjects. Intervals resample subjects and repeat the whole procedure, calibration included. Some subjects may defer, so the forced-choice subgroup is reported separately: those subjects cannot defer, and no coding decision about deferral enters that row.

Subjects	n	Choices	Renorm.	Race	Gain
All	364	8,239	0.9263	0.9164	+0.0100 [+0.0048, +0.0246]
Forced choice only	137	3,425	0.9112	0.8846	+0.0265 [+0.0116, +0.0700]

Table 1: Held-out log loss per choice on menus of size two to four, with both rules calibrated only to training-fold full-menu shares under a common add-α convention, $\alpha = 1/2$. Shares rest on 268 full-menu choices for all subjects and 109 for the forced-choice subgroup. Intervals are subject bootstraps that recompute folds, shares, calibration and predictions.

The Gaussian rule is better on both, and better by more where deferral cannot interfere. Section 4 shows the gain is not an artefact of estimating shares from a few hundred observations.

3 Choices induced from rankings

Twelve archival collections have human respondents and rankings, or ratings fine enough to yield them. Two rank-order card tasks from the General Social Survey ask what a child should learn and what matters in a job. Three independent samples of Jester users rate the same ten jokes. Sushi rankings come from a Japanese web survey, film rankings are induced from Netflix ratings, and dots and puzzles are perceptual discrimination tasks. The rest are rankings of seven sports by participation preference, ten occupations by perceived prestige, and four political goals from a five-nation survey. The thirteen collections comprise 147,719 responses, of which the menu experiment contributes 373 subjects; the twelve ranking collections contribute 147,346. These are not thirteen independent experiments, since three are Jester files, two are General Social Survey items, and dots and puzzles are companion tasks.

The protocol matches Section 2 with respondents in place of subjects, and the outcome for a subset is the respondent's highest-ranked member of it.

Dataset	n	K	Subsets	Renorm.	Race	Gain
Occupational prestige	143	10	1,013	1.0173	0.9783	+0.0390 [+0.0227, +0.0555]
Sushi	5,000	10	1,013	1.3724	1.3613	+0.0111 [+0.0094, +0.0128]
Political goals	2,262	4	11	0.8835	0.8766	+0.0069 [+0.0059, +0.0078]
GSS socialization	5,000	5	26	0.7984	0.7929	+0.0055 [+0.0041, +0.0070]
Jester file 3	5,000	10	1,013	1.5189	1.5165	+0.0024 [+0.0015, +0.0032]
Jester file 1	5,000	10	1,013	1.5287	1.5267	+0.0020 [+0.0012, +0.0027]
Jester file 2	5,000	10	1,013	1.5247	1.5230	+0.0018 [+0.0010, +0.0025]
GSS job values	5,000	5	26	0.8593	0.8576	+0.0017 [+0.0005, +0.0030]
Dots	3,183	4	11	0.8285	0.8270	+0.0015 [+0.0005, +0.0025]
Netflix	4,379	3	4	0.7932	0.7922	+0.0009 [+0.0007, +0.0012]
Puzzles	3,180	4	11	0.8180	0.8173	+0.0007 [−0.0003, +0.0017]
Sports participation	130	7	120	1.2579	1.2528	+0.0050 [+0.0017, +0.0087]

Table 2: Held-out log loss per prediction over every subset of size two or more. Datasets are capped at five thousand respondents for runtime. The race is lower in twelve of twelve and the interval excludes zero in eleven. Occupational prestige is scorable here only because the shared smoothing removes a zero training cell. Sports participation sits below the rule: its gain is real but Section 4 shows it carries no evidence about the restriction map. Intervals resample respondent losses with the fitted training models held fixed, so unlike Table 1 they omit calibration uncertainty and are too narrow, most seriously in the two smallest rows.

Averaging over every subset weights cardinalities by how many there are, so intermediate menus dominate: for $K = 10$ only 45 of 1,013 subsets are pairs, about four per cent of the weight. Those middling menus are where least has been removed and least is at stake. Splitting by menu size shows the pairwise gain exceeding the aggregate in eleven of twelve datasets, by factors from 0.7 to 3.7 with median 1.8: Sushi is +0.0412 at $|T| = 2$ against +0.0111 aggregate, and GSS socialization +0.0130 against +0.0055. Occupational prestige is the exception, peaking at $|T| = 4$. At $|T| = K$ nothing has been removed, the two rules must agree, and the measured gain is zero to four decimals in every dataset, which is the check that the pipeline is sound. We keep the all-subsets aggregate as the headline because it fixes the estimand in advance rather than selecting a cardinality after the fact.

4 Shrinkage

A positive gain has two possible sources and log loss cannot tell them apart. The race may be closer to how the population chooses. Or the race may be wrong about that and win anyway, because contracting a noisily estimated share vector is shrinkage, and shrinkage lowers log loss when the input is over-confident. The second mechanism needs no departure from the axiom, so properness of the score licenses nothing here: a proper score says the true probability vector is optimal, not that a plug-in estimate of it beats a biased transformation of the same estimate.

The remedy is to build a population where the axiom holds by construction and run the identical procedure on it. Taking the observed full-menu shares as worths, choices or complete rankings are generated from that exact process, and everything is repeated. Rankings are drawn by sorting $\log u_i + G_i$ for independent standard Gumbel G_i, which is the Plackett–Luce ranking law; for $K \geq 4$ this is a particular joint law consistent with those subset marginals rather than the only one, so the null is specifically a Plackett–Luce ranking null.

Dataset	Observed	Null median	Excess	MC tail
Menu experiment, forced choice	+0.0265	−0.0041	+0.0306	0.010
Menu experiment, all subjects	+0.0100	−0.0023	+0.0123	0.015
Occupational prestige	+0.0390	−0.0227	+0.0617	$\leq$ 0.005
Sushi	+0.0111	−0.0062	+0.0174	$\leq$ 0.005
GSS socialization	+0.0055	−0.0066	+0.0121	$\leq$ 0.005
Political goals	+0.0069	−0.0012	+0.0081	$\leq$ 0.005
GSS job values	+0.0017	−0.0047	+0.0065	$\leq$ 0.005
Jester file 3	+0.0024	−0.0015	+0.0039	$\leq$ 0.005
Jester file 1	+0.0020	−0.0014	+0.0034	$\leq$ 0.005
Jester file 2	+0.0018	−0.0014	+0.0032	$\leq$ 0.005
Dots	+0.0015	−0.0014	+0.0029	$\leq$ 0.005
Puzzles	+0.0007	−0.0011	+0.0019	$\leq$ 0.005
Netflix	+0.0009	−0.0001	+0.0010	$\leq$ 0.005
Sports participation	+0.0050	+0.0051	−0.0000	0.51

Table 3: The same gain when the axiom holds by construction. Two hundred replicates per row. The diagnostic is the excess of the observed gain over the null median; the tail probabilities are indicative rather than exact tests, and for the menu rows they are optimistic, because the null redraws each subject's thirty-one choices independently given the aggregate worths and so omits the within-subject dependence that the bootstrap intervals of Table 1 do preserve. Wherever shares rest on more than about two thousand observations the null is negative, because contraction is then a bias rather than useful insurance, so comparing an observed gain with zero understates it.

The shrinkage mechanism is real and it accounts for the entire gain in the smallest dataset. Sports participation, with 130 respondents, sits exactly on its null and is excluded from every claim here. Everywhere else the mechanism runs against the race, and the excess over the null exceeds the raw gain in all twelve remaining rows. It also rescues puzzles, whose raw interval in Table 2 covers zero while its excess over the null does not. The menu experiment clears the null on both readings, which is the result the paper rests on, since it is the only one where restriction is observed.

5 The benchmark gap

Most of the aggregate gains in Tables 1 and 2 are a few thousandths of a nat, and such a figure is hard to read on its own. Neither map is permitted to see a restricted menu, and most of what is predictable about a restricted menu is already contained in the full-menu probabilities that both of them receive.

We therefore score a third forecast which is permitted to see them. Fix a subset T and a training fold. Among the training respondents, count how often each item of T was the one taken from T, smooth those counts as above, and use them as the forecast for held-out respondents facing T. This is a direct estimate of the choice probabilities within T, built from exactly the observations the two maps are denied, and it is scored on the same held-out respondents by the same log loss.

It beats both maps, as it should. The question is by how much, because that difference is the room available. Define the gap as the held-out loss reduction of this counting forecast relative to renormalization, and read the race's share of that gap from Table 4.

Dataset	Gap	Captured	Corrected gap	Captured
Dots	0.0023	65%	0.0026	58%
Jester file 1	0.0041	49%	0.0046	43%
Jester file 3	0.0051	47%	0.0056	43%
Political goals	0.0150	46%	0.0154	45%
Jester file 2	0.0040	45%	0.0045	40%
Sushi	0.0260	43%	0.0265	42%
Occupational prestige	0.1135	34%	0.1312	30%
Puzzles	0.0026	27%	0.0029	24%
GSS socialization	0.0223	25%	0.0225	24%
Netflix	0.0046	20%	0.0048	19%
GSS job values	0.0087	20%	0.0089	19%
Median		43%		40%

Table 4: Gap is the held-out loss reduction of a smoothed per-subset frequency forecast relative to renormalization. Captured is the race's share of it. The eleven are the ranking collections whose observed gain exceeds the corresponding null simulation in Table 3; sports participation is excluded throughout. Renormalization captures none of the gap, being the unique map that preserves every ratio.

That third forecast is not an oracle. It is estimated too, so its expected held-out loss exceeds the true conditional entropy by approximately $(m-1)/2n$ for an m-outcome menu and n training observations. The last two columns add that correction to the gap, which lowers the median from 43 to 40 per cent.

Two of the gaps are near 0.002, so the dots and puzzles ratios are the least stable. The percentages are context for the absolute gains.

6 Scope

The practical recommendation is narrow. When the task is to infer restricted-menu population shares from full-menu population shares, the Gaussian race should be the default in place of renormalization. It requires only a low-dimensional numerical calibration and one-dimensional quadrature, needs no additional data, and was better on every collection examined here except the one too small to discriminate. In the single experiment where restriction is actually observed rather than induced from rankings, it is better by +0.0265 nats per choice among subjects who cannot defer.

Ten of the twelve ranking collections carry an assumption the menu experiment does not. A respondent's ranking imposes one ordering on every subset by construction, so no ranking dataset can show a person choosing i over j from one menu and j over i from another, and none reveals whether that person would choose as their ranking implies when handed two options. Reading restriction off a ranking assumes they would. Section 2 does not.

Every quantity here is a population quantity. A population of people who each choose by Equation (1), with worths differing between people, generally does not satisfy the axiom in aggregate, which is why mixed logit is useful and why its probabilities are integrals over logit probabilities rather than logit probabilities [7]. Aggregate failure of the axiom is therefore not evidence that individuals violate it. Heterogeneous Luce choosers reproduce aggregate contraction. What is rejected is a representative-agent map.

The experiment compares two restriction maps, not two fitted likelihoods. A fitted Plackett–Luce model uses the whole ranking and a fitted Bradley–Terry model uses the pairwise outcomes directly. Neither was estimated here. Recommendations about ranking or paired-comparison likelihoods therefore have no support in these tables.

The score is a sum of marginal scores for top-item-within-subset outcomes. It is proper for those marginals and not for the ranking distribution. With $K = 4$ the scored marginals carry at most 17 degrees of freedom against 23 for a distribution over rankings, so distinct ranking laws can agree on everything scored here.

Nesting is not costlessness. Conceptually the independent-noise race family nests Luce at the Gumbel noise law, so the family excludes no Luce population, but Table 3 shows that choosing the Gaussian point is a separate statistical decision which loses when the axiom holds and the shares are well estimated.

So the recommendation is not a claim of dominance. Neither map dominates the other: when the shares are precise the better map is whichever one matches the population, and when they are very noisy the dominant consideration is estimation risk rather than either structural rule. What the tables support is a structural default in the absence of evidence for the axiom, which is the situation an analyst holding only aggregate shares is actually in.

A tempting shortcut fails. Pricing ordered outcomes by removing the winner and re-running looks like the race's own prediction, and it is not: against the calibrated race's ordering law it misprices individual pairs and triples by up to a factor of 3.4. We leave to future work the comparison of the exact Gaussian ordering law against Plackett–Luce ranking probabilities on complete-ranking data.

Data and code

Sushi rankings come from Kamishima [4]; Jester ratings from the Eigentaste project [3]; the General Social Survey cumulative file from NORC; the perceptual, leisure, prestige, film and political-goal rankings via PrefLib [6] and the CRAN packages PLMIX and ConsRank; the menu-choice data from the replication archive of [1]. All are public. Analysis code and every input needed to reproduce the tables are under `research/human` in `github.com/microprediction/winning`, the inputs as column extracts rather than raw archives.

Two storage conventions must be told apart when reproducing this. BayesMallows, ConsRank and PerMallows store ranks, so element j holds the rank of alternative j, while PLMIX stores orderings, so the first element names the favourite. Reading one as the other transposes the data silently and raises no error. We established which is which by cross-validation: the political goals data ships in two of these packages and they agree only when each is read under its own convention.

A Why a population should look Gaussian

Nothing so far explains why the Gaussian member of the class should be the one that fits. A mechanism is available, and it is elementary.

Let every individual obey the axiom, and let their tastes differ.

Proposition 2 (A population of Luce choosers is a contest)**.** *Suppose individual θ chooses by Luce with log-worths $\mu_i + \sigma Z_i(\theta)$, where $Z(\theta) \sim N(0, I_K)$ independently across individuals, and with Gumbel choice noise of scale β. Then the population choice probabilities are exactly those of a contest in which alternative i draws*

$$\mu_i + \sigma Z_i + \beta G_i, \tag{4}$$

independent across alternatives. Writing $\mu_i = \sigma a_i$ and letting $\sigma/\beta \to \infty$, those probabilities converge to the independent unit-variance Gaussian race of Equation (2) with locations a.

Proof. By Yellott's theorem individual θ chooses $\arg\max_i(\mu_i + \sigma Z_i(\theta) + \beta G_i(\theta))$. Marginalising over θ leaves the per-alternative noise $\sigma Z_i + \beta G_i$, independent across i because both components are, which gives (4) exactly and at every σ. Dividing by σ preserves the $\arg\max$ and sends the Gumbel term to zero, and ties have probability zero in the limit, so continuous mapping transfers convergence to the winning probabilities. □

The proposition is exact at every σ, and only the limit is asymptotic. Two remarks. The location scaling is not decoration: holding μ fixed while σ grows sends every share to $1/K$, so the Gaussian race would be recovered only at near-indifference. And the Gaussian assumption on taste is where a central limit theorem belongs: if an individual's worth for an alternative accumulates many small independent influences, taste variation across a population is approximately Gaussian for the usual reason. The Gumbel plays no part in the limit, only in making each individual exactly Luce, so any finite-variance taste law with a Gaussian limit does the same work. Thurstone's race is what aggregation delivers, and Luce sits in its basin of attraction.

σ	Top share	Observed λ	Case V λ	Ratio
0.5	0.255	0.052	0.225	0.23
1	0.300	0.128	0.227	0.57
2	0.343	0.197	0.229	0.86
3	0.359	0.216	0.229	0.94
4	0.365	0.228	0.230	0.99
8	0.373	0.230	0.231	1.00
16	0.375	0.233	0.231	1.01

Table 5: A population of Luce choosers whose log-worths carry Gaussian taste variation of scale σ against a Gumbel choice-noise scale of 1, with locations scaled as σa so that the shares do not degenerate. Observed λ is the population's contraction; Case V λ is what the independent unit-variance Gaussian race implies for that population's own shares. The top share is reported to show the limit is not the near-indifference corner.

The consequence for this paper is that no individual need be non-Lucean for the Gaussian transport to be the right one upstairs. A population of Luce choosers whose tastes vary is a contest with composite noise, and once taste variation outweighs within-person choice noise the composite is Gaussian and the population contracts by just what Case V predicts. The aggregate caveat of Section 6 is therefore the mechanism rather than an apology, and it predicts where the fit should be worst: in populations whose members are nearly alike.

Independent evidence for the same phenomenon exists in memory research. Robinson and colleagues [8] compare softmax against signal detection across m-alternative tasks and report that the softmax parameter must vary with m while the signal-detection parameter is stable. That is the contraction studied here, seen through the parameter that has to move to absorb it.

References

[1] Miguel A. Costa-Gomes, Carlos Cueva, Georgios Gerasimou, and Matúš Tejivščák. Choice, deferral and consistency. *Quantitative Economics*, 13:1297–1318, 2022.

[2] Peter Cotton. Inferring relative ability from winning probability in multientrant contests. *SIAM Journal on Financial Mathematics*, 12(1), 2021.

[3] Ken Goldberg, Theresa Roeder, Dhruv Gupta, and Chris Perkins. Eigentaste: a constant time collaborative filtering algorithm. *Information Retrieval*, 4(2):133–151, 2001.

[4] Toshihiro Kamishima. Nantonac collaborative filtering: recommendation based on order responses. In *Proceedings of KDD*, 2003.

[5] R. Duncan Luce. *Individual Choice Behavior: A Theoretical Analysis*. Wiley, 1959.

[6] Nicholas Mattei and Toby Walsh. PrefLib: a library of preference data. In *Algorithmic Decision Theory*, 2013.

[7] Daniel McFadden and Kenneth Train. Mixed MNL models for discrete response. *Journal of Applied Econometrics*, 15(5):447–470, 2000.

[8] Maria M. Robinson, Isabella C. DeStefano, Edward Vul, and Timothy F. Brady. The relationship between Luce's choice axiom and signal detection theory in m-alternative tasks. *Journal of Mathematical Psychology*, 117:102805, 2023.

[9] Louis L. Thurstone. A law of comparative judgment. *Psychological Review*, 34:273–286, 1927.

[10] John I. Yellott. The relationship between Luce's choice axiom, Thurstone's theory of comparative judgment, and the double exponential distribution. *Journal of Mathematical Psychology*, 15:109–144, 1977.